

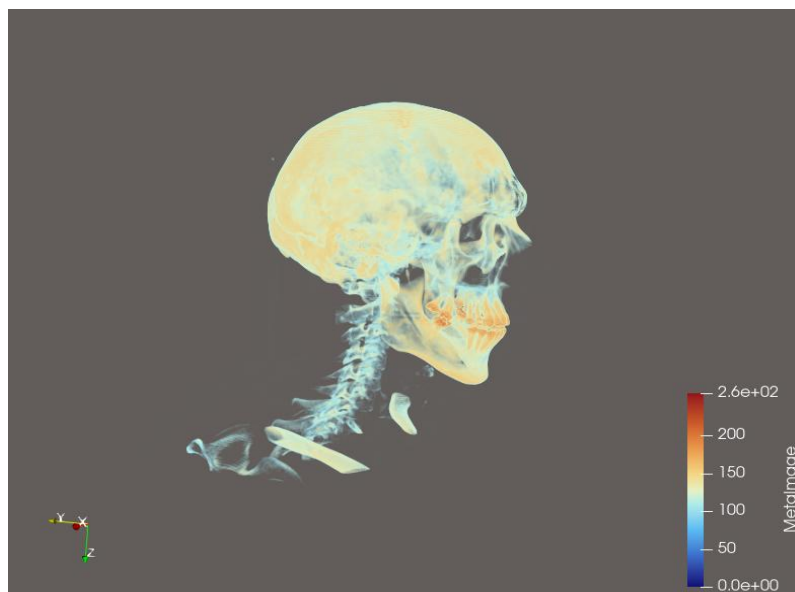
Part 1: Visualization with ParaView

Introduction

In this assignment, ParaView was used to visualize different types of scientific datasets using multiple visualization techniques, including volume rendering, surface rendering, streamlines, and molecular visualization. The objective was to understand how visualization methods depend on the type of data being analysed.

(a) VisHuman Head – Volume Rendering

The VisHuman Head dataset was visualized using **direct volume rendering** in ParaView. Since this is a 3D scalar volume dataset, a transfer function was applied to map intensity values to colours and opacity. Low-density regions were made transparent, while high-density regions such as bone were rendered more opaque and brighter. This allowed internal anatomical structures to be clearly visible without extracting surfaces.



(b) Asian Dragon – Surface Mesh Visualization

The Asian Dragon dataset downloaded from the Stanford 3D Scanning Repository was loaded into ParaView and visualized using **surface rendering**. This dataset consists of a **3D polygonal mesh**, not volumetric data. As a result, techniques such as volume rendering or transfer functions are not applicable. The visualization highlights detailed geometric features of the dragon model.



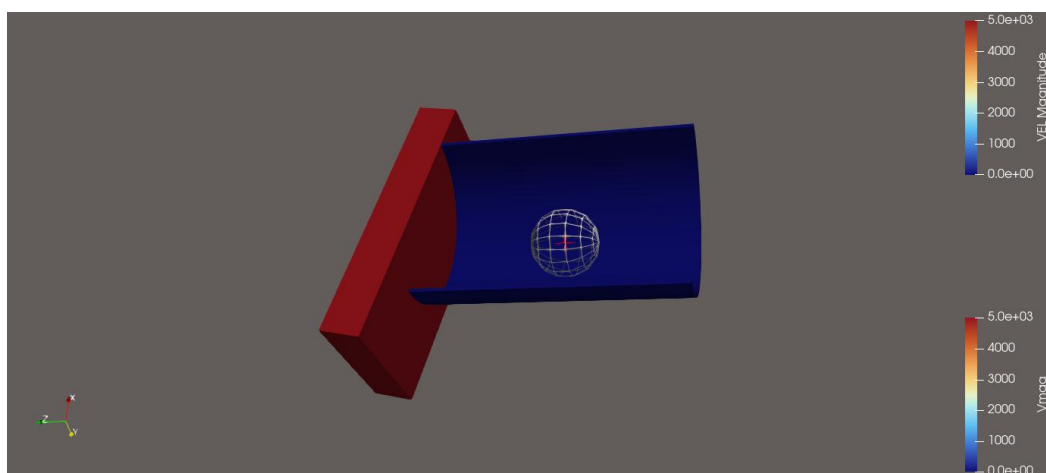
(c) Vector Data – Streamlines and Particle Tracing

For vector field visualization, a **vector dataset** was used containing velocity information (Vmag). This dataset represents a **3D vector field**, commonly found in fluid flow or simulation data.

The following visualization techniques were applied:

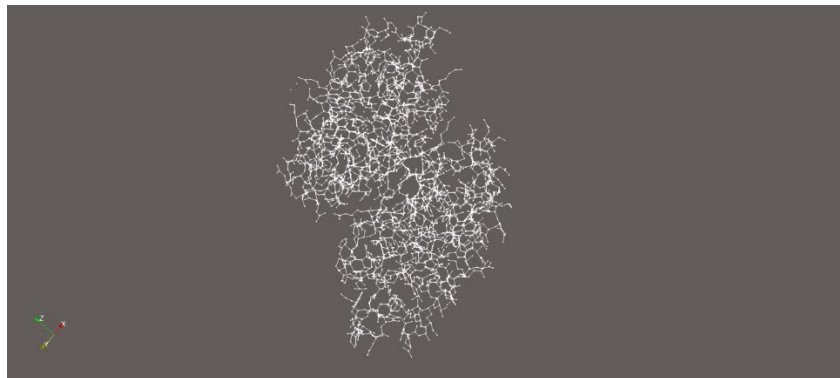
- Stream Tracer filter to generate streamlines that follow the vector field direction
- Colouring based on velocity magnitude (Vmag)
- Seeding of particles at selected locations to observe flow behaviour

The resulting visualization clearly shows how particles move through the domain and how velocity magnitude varies spatially. High-velocity regions are highlighted using warmer colours, while low-velocity regions appear cooler. This visualization demonstrates ParaView's effectiveness for analysing flow patterns and vector-based simulation data.

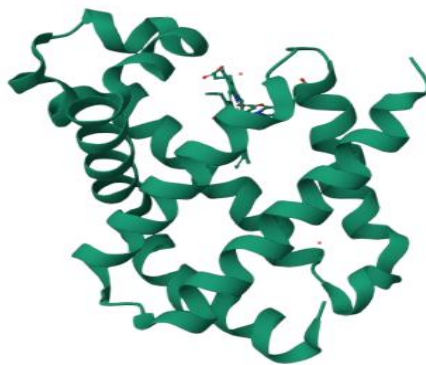


(d) Protein Structures – PDB Visualization and Comparison

Several protein structures files (.pdb) were downloaded from the RCSB Protein Data Bank and visualized in ParaView. The protein geometry and bonds were visible, but ParaView provides limited molecular-specific visualization options. The same proteins were also viewed using the RCSB online 3D viewer, which offers more specialized molecular representations. This comparison highlights the difference between a general-purpose visualization tool and a domain-specific one.



- Protein structure visualized in ParaView



- Same protein visualized in the RCSB online 3D viewer

Part 2- Superstore Sales Analysis Using Tableau

Introduction

This report presents an exploratory data analysis of the *Sample Superstore* dataset using **Tableau**. The objective of this assignment is to visualize and analyze sales and profit performance across different states and over time, and to compare the performance of selected states using interactive filters.

To achieve this, several visualizations were created and combined into a single interactive dashboard. These include a geographic map of sales by state, a time series of sales over months, and a scatter plot showing the relationship between profit and sales. The analysis is structured according to parts (a) through (d) of the assignment.

Part (a): Sales Distribution by State

To analyse how sales are distributed geographically, a filled map visualization was created showing total sales by U.S. state.

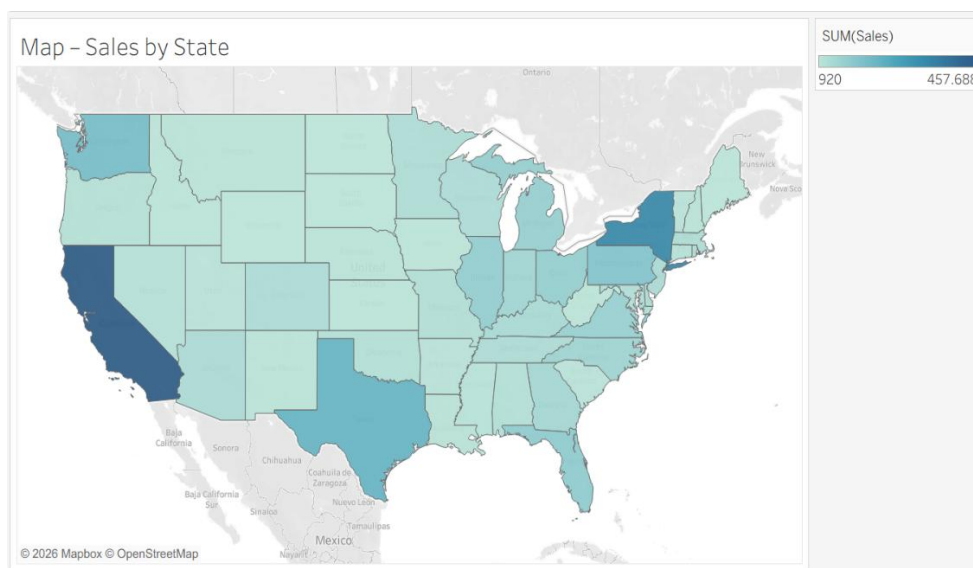
Visualization used: Map – Sales by State

Measure: SUM(Sales)

Dimension: State/Province

Each state is color-coded based on its total sales value, where darker shades indicate higher sales. From this visualization, it is evident that states such as **California, New York, and Texas** contribute significantly higher sales compared to many other states. In contrast, several central and smaller states show relatively lower sales values.

This map provides a quick and intuitive overview of regional sales performance across the United States.



Part (b): Sales Trends Over Time of Geographically Close States

Visualization used: *Sales Over Time*

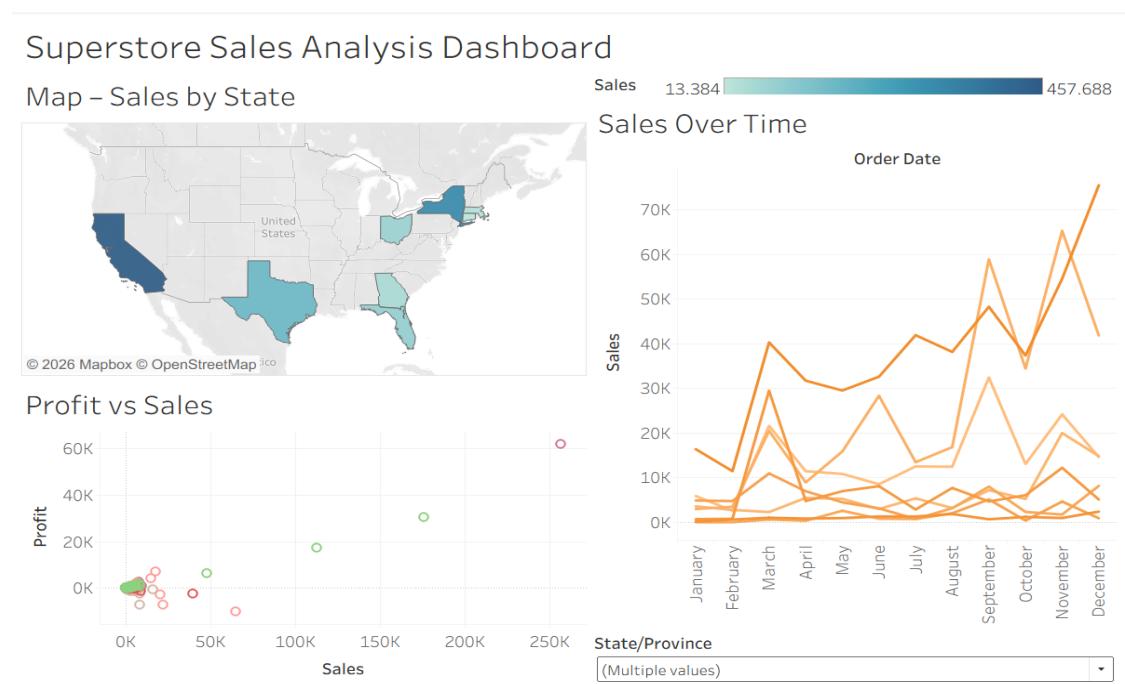
Measure: SUM(Sales)

Dimension: Order Date (Month)

Colour: State/Province

To analyse whether geographically close states show similar sales behaviour over time, the Sales Over Time line chart was used together with the geographic map. By selecting neighbouring states on the map (for example, states within the same region), their corresponding sales trends over time are highlighted in the line chart.

It is observed that some geographically close states exhibit similar seasonal patterns, such as increased sales toward the end of the year, while others still differ in magnitude. This suggests that geographic proximity can lead to similar temporal trends, although total sales volumes may vary between states.



Part (c): Relationship Between Profit and Sales

To examine the relationship between sales and profitability, a **scatter plot** was created plotting total profit against total sales.

Visualization used: *Profit vs Sales*

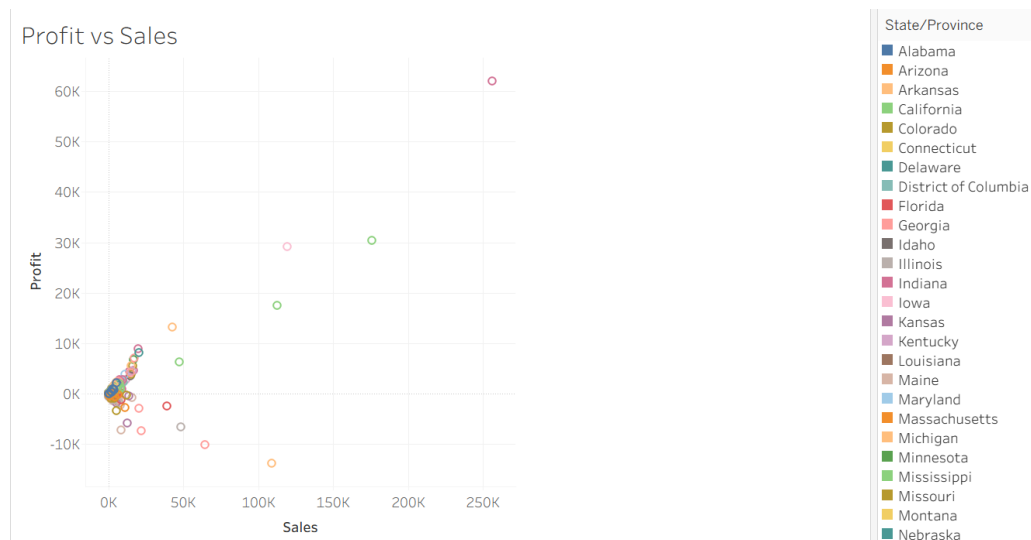
X-axis: SUM(Sales)

Y-axis: SUM(Profit)

Colour: State/Province

This scatter plot reveals a general positive relationship between sales and profit—states with higher sales tend to generate higher profits. However, the relationship is not perfectly linear. Some states show moderate sales but negative or very low profit, indicating inefficiencies such as high discounts or operational costs.

Outliers can also be observed where certain states achieve high profits relative to their sales, suggesting stronger margins or more efficient operations.



Part (d): Comparison of Florida with Other States

To compare the performance of **Florida** with other states, the **State/Province filter** was applied and set to *Multiple Values*. Florida was selected along with a small number of other major states (e.g., California, Texas, and New York). This selection was applied across the entire dashboard to allow direct comparison.

Visualization used: *Interactive Dashboard with State Filter*

From the dashboard view, Florida's performance can be clearly compared with other states:

- Florida cities generally exhibit **moderate sales levels**

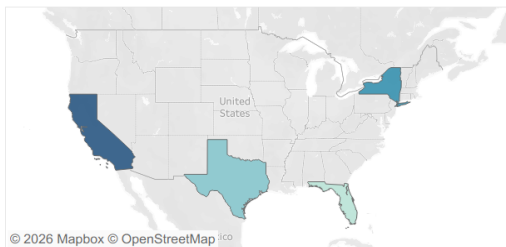
- Profit values for Florida cities are **mixed**, with several points close to zero or negative profit
- At similar sales levels, Florida tends to have **lower profit margins** compared to states such as California or New York

This comparison indicates that although Florida generates reasonable sales, its profitability is relatively weaker, suggesting potential issues related to costs, discounts, or product mix.

Using multiple selected states is essential here, as selecting only Florida would remove the comparative context. The dashboard approach allows all charts (map, time series, and scatter plot) to update simultaneously, making comparison intuitive and effective.

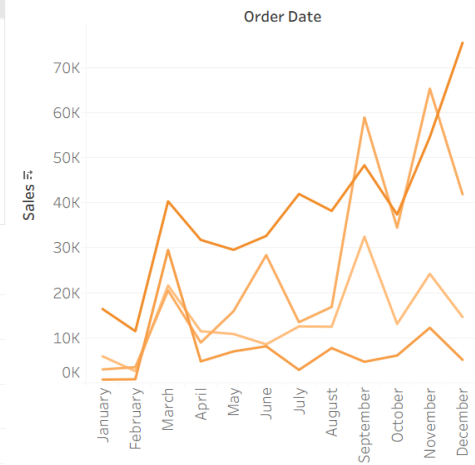
Superstore Sales Analysis Dashboard

Map – Sales by State

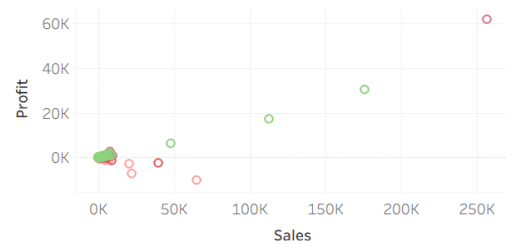


Sales 89,474 457,688

Sales Over Time



Profit vs Sales



State/Province
(Multiple values)

Conclusion

In this assignment, Tableau was used to effectively analyze and visualize sales and profit data from the Superstore dataset. The geographic map highlighted regional sales differences, the time series revealed seasonal sales patterns, and the scatter plot illustrated the relationship between sales and profitability.

The interactive dashboard enabled meaningful comparisons between states, particularly highlighting Florida's moderate sales performance and relatively lower profit margins compared to other major states. Overall, the analysis demonstrates how Tableau dashboards can be used to gain actionable business insights through visual analytics.